

Lec 13-1, Dec 1.

- Solving first-order D.E.
 - Separable equations (7.4)
 - linear equations (9.2)
- Applications of D.E. (9.3)
 - Growth models
 - Newton's law of cooling
 - Mixture problems
 - Orthogonal trajectories

Solving First-order D.E.

• Separable equations

Def: A D.E. of the form $\frac{dy}{dx} = g(x)f(y)$ is said to be separable.

e.g. $y' = e^{x+y}$ is separable, while $y' = e^x + e^y$ is not.
 $y' = e^x \cdot e^y = g(x) \cdot f(y)$ $g(x)f(y)$

Suppose that $y' = g(x)f(y)$ and f is not the zero function. Then

$$\frac{1}{f(y)} y' = g(x)$$

$$\int dx \text{ both sides} \implies \int \frac{1}{f(y)} \underbrace{y' dx}_{\frac{dy}{dx} \cdot dx} = \int g(x) dx$$

$$\implies \int \frac{1}{f(y)} dy = \int g(x) dx$$

$$\implies F(y) = C + \int g(x) dx \implies y = F^{-1}(\int g(x) dx + C) \quad \square$$

After integration, y is written as an implicit function of x , and sometimes we may solve y explicitly in terms of x .

Shortcut If $\frac{dy}{dx} = g(x)f(y)$, then " $\frac{dy}{f(y)} = g(x)dx$ "

and " $\int \frac{1}{f(y)} dy = \int g(x) dx$ "

e.g.1. Solve $y' = e^{x+y}$.

Solⁿ: $\frac{dy}{dx} = e^x e^y \Rightarrow e^{-y} dy = e^x dx$

$$\Rightarrow \int e^{-y} dy = \int e^x dx$$

$$\Rightarrow -e^{-y} = e^x + C \quad (\text{where } C < 0)$$

$$\Rightarrow e^{-y} = -e^x - C$$

$$\Rightarrow y = -\ln(-e^x - C),$$

where C is any appropriate real number:

And the domain for a given $C < 0$ is

$$D = \{x \in \mathbb{R} : -e^x - C > 0\} = (-\infty, \ln(-C)) \quad \square$$

check: $y' = -\frac{1}{-e^x - C} (-e^x) = -\frac{e^x}{e^x + C}$

$$e^x \cdot e^y = e^{-\ln(-e^x - C)} = \frac{1}{-e^x - C} \cdot e^x = -\frac{e^x}{e^x + C} \Rightarrow y' = e^y \cdot e^x.$$

e.g.2. Solve $y' = x^2 y$.

Sol': . Note that $y \equiv 0$ (zero function) is a solution .

· Suppose y is not the zero function. . Then

$$\frac{dy}{dx} = x^2 y \Rightarrow \int \frac{1}{y} dy = \int x^2 dx$$

$$\Rightarrow \ln|y| = \frac{1}{3}x^3 + C_1$$

$$\Rightarrow |y| = e^{\frac{1}{3}x^3 + C_1}, \quad C_1 \text{ is any real number}$$

$$\begin{aligned} \Rightarrow y &= \pm e^{\frac{1}{3}x^3 + C_1} \\ &= \pm e^{\frac{1}{3}x^3} \cdot C_2 \quad (\text{where } C_2 = e^{C_1} > 0) \\ &= e^{\frac{1}{3}x^3} \cdot C_3 \quad (\text{where } C_3 \text{ is any nonzero constant}) \end{aligned}$$

where C_2 is a positive const' & C_3 is a nonzero const'.

· Combining with the zero function (also a solution), we have

the general solution

$$y(x) = e^{\frac{1}{3}x^3} \cdot C, \quad \text{where } C \text{ is any real number. } \quad (1)$$

check: y satisfies the D.E.

$$\begin{aligned} y' &= C \cdot \frac{1}{3} e^{\frac{1}{3}x^3} \cdot 3x^2 = x^2 e^{\frac{1}{3}x^3} \cdot C \quad \text{for any fixed } C \in \mathbb{R} \\ &= x^2 \cdot y \end{aligned}$$

e.g. 3. Solve the IVP $y' = (\frac{x}{y})^2$, $y(0) = 2$.
initial value problem

Sol': $\frac{dy}{dx} = \frac{x^2}{y^2} \Rightarrow y^2 dy = x^2 dx$

$$\Rightarrow \frac{1}{3} y^3 = \frac{1}{3} x^3 + C$$

$$\Rightarrow y = (x^3 + 3C)^{\frac{1}{3}}, \text{ where } C \text{ is any const.}$$

Compute: $y(0) = (x^3 + 3C)^{\frac{1}{3}} \Big|_{x=0} = (3C)^{\frac{1}{3}} = 2 \Rightarrow (3C)^{\frac{1}{3}} = 2 \Rightarrow 3C = 8.$

$\therefore$ The (particular) solution is $y = (x^3 + 8)^{\frac{1}{3}}$. □.

Linear equations

Def: A first-order linear D.E. is an equation of the form

$$\frac{dy}{dx} + P(x) \cdot y = Q(x). \quad \text{[or: } \frac{dy}{dx} = -P(x)y + Q(x) \text{].}$$

Rmk. General linear DEs have the form

$$P_0(x)y + P_1(x)y' + \dots + P_n(x) \cdot y^{(n)} = Q(x), \quad \text{[no need to remember the general form]}$$

where $P_i(x)$ and $Q(x)$ do not have to be linear.

Q: How do we solve a linear D.E.?

$$y' = \left(-\frac{1}{x}\right)y + 2$$

Consider solving $y' + \frac{y}{x} = 2$ for $x > 0$, which is linear.

multiplying both sides by x yields

$$x \left(y' + \frac{y}{x} \right) = 2x \Rightarrow xy' + y = 2x$$

$$\Rightarrow (x \cdot y)' = 2x$$

[apply $\int dx$ both sides]

$$\Rightarrow xy = \int 2x dx = x^2 + C$$

$$\Rightarrow y = \frac{1}{x}(x^2 + C), \text{ where } C \in \mathbb{R}.$$

This example above becomes easy once we realized that the left-hand side is the derivative of some function $v(x)y$.

Consider solving $y' + P(x)y = Q(x)$. Consider multiplying both sides by $v(x)$ (where v is not zero function):

$$(v(x)y)' = v(x)y' + v(x)P(x)y = v(x)Q(x).$$

key idea: make LHS be a derivative!

If we can choose $v(x)$ s.t. the left-hand side is $(v(x)y)'$,
then $(v(x)y)' = v(x)Q(x)$.

$$\Rightarrow v(x)y = \int v(x)Q(x) dx.$$

Symbolically, we can denote this by

$$y = \frac{1}{V(x)} \int V(x) Q(x) dx$$

this will contain "+C".

Any function $V(x)$ (not identically zero) that satisfies

$$\frac{d(V(x)y)}{dx} = V(x) \left(\frac{dy}{dx} + P(x)y \right)$$

is called an integrating factor.

Q: How can one find an integrating factor?

A: Exponential! $V(x) = \exp\left\{\int p(x) dx\right\}$

check:

$$\begin{aligned} \frac{d(\exp\{\int p(x) dx\} y(x))}{dx} &= e^{\int p(x) dx} p(x) y(x) + e^{\int p(x) dx} y'(x) \\ &= e^{\int p(x) dx} [p(x) y(x) + y'(x)] \\ &= V(x) [p(x) y(x) + y'(x)] \end{aligned}$$

To solve $y' + P(x)y = Q(x)$, follow the steps below:

1. Let $v(x) := e^{\int P(x)dx}$ (an integrating factor), where $\int P(x)dx$ is any antiderivative of $P(x)$.
2. The general solution is given by

$$y = \frac{1}{v(x)} \int v(x)Q(x) dx.$$

Explain. $y' + P(x)y = Q(x) \xRightarrow{\text{key step}} e^{\int P(x)dx} (y' + P(x)y) = e^{\int P(x)dx} \cdot Q(x)$

make LHS be a derivative

$$\Rightarrow [e^{\int P(x)dx} y]' = e^{\int P(x)dx} \cdot Q(x)$$

$$\Rightarrow e^{\int P(x)dx} y = \int e^{\int P(x)dx} \cdot Q(x) dx$$

$$\Rightarrow y(x) = \frac{1}{e^{\int P(x)dx}} \int e^{\int P(x)dx} \cdot Q(x) dx. \quad \square$$

e.g.4 Solve the IVP

$$\begin{cases} xy' + 2y = x^2 - x + 1, & x > 0. \\ y(1) = \frac{1}{2}. \end{cases}$$

↓

Sol' : draft :

$$y' = \left(-\frac{2}{x}\right)y + \left[x - 1 + \frac{1}{x}\right]$$

Step 1: standard form!

We first have

$$y' = \left(-\frac{2}{x}\right)y + \left[x - 1 + \frac{1}{x}\right]$$

Step 2: Integrating factor! Exponential!

$$\text{Let } V(x) = \exp\left\{\int \left(-\frac{2}{x}\right) dx\right\} = \exp\{2 \ln|x|\} = x^2.$$

$$\therefore x^2 y' = x^2 \left[\left(-\frac{2}{x}\right)y + \left[x - 1 + \frac{1}{x}\right] \right]$$

$$x^2 y' = -2xy + x^2 \left(x - 1 + \frac{1}{x}\right)$$

$$(x^2 \cdot y)' = x^2 y' + 2xy = x^2 \left(x - 1 + \frac{1}{x}\right)$$

Step 3: Solve out $y(x)$!

$$\therefore x^2 \cdot y = \int x^2 \left(x - 1 + \frac{1}{x}\right) dx = \int (x^3 - x^2 + x) dx$$

$$\begin{aligned} \therefore y(x) &= \frac{1}{x^2} \left[\frac{1}{4} x^4 - \frac{1}{3} x^3 + \frac{1}{2} x^2 + C \right] \\ &= \frac{1}{4} x^2 - \frac{1}{3} x + \frac{1}{2} + \frac{1}{x^2} C, \text{ where } C \text{ is any real number.} \end{aligned}$$

Step 4: Substitute initial condition!

$$\therefore y(1) = \left[\frac{1}{4} x^2 - \frac{1}{3} x + \frac{1}{2} + \frac{1}{x^2} C \right] \Big|_{x=1} = \frac{1}{2} \Rightarrow C = \frac{1}{12}.$$

$$\therefore y(x) = \frac{1}{4} x^2 - \frac{1}{3} x + \frac{1}{2} + \frac{1}{x^2} \frac{1}{12}.$$

□

check by yourself (*) satisfies all the conditions.

e.g.5. Solve the D.E.

$$\cos(x) \cdot y' + \sin(x) \cdot y = 2\cos^3(x) \sin(x) - 1, \quad 0 \leq x < \frac{\pi}{2}.$$

Sol':

draft:

$$y' = -(\tan x) y + \frac{2\cos^3 x \sin x - 1}{\cos x}$$

$$\text{Let } v(x) = \exp \left\{ \int \tan x \, dx \right\} = \exp \left\{ \ln |\sec x| \right\} = |\sec x| = \sec x \quad (\because x \in (0, \frac{\pi}{2}))$$

$$\Rightarrow v(x) y' = v(x) [-(\tan x) y + 2\cos^2 x \sin x - \sec x]$$

$$\Rightarrow v(x) y' + v(x) \tan x y = v(x) \cdot (2\cos^2 x \sin x - \sec x)$$

$$\Rightarrow (e^{\int \tan x \, dx} y)' = v(x) (2\cos^2 x \sin x - \sec x)$$

$$\Rightarrow (v(x) y(x))' = (\sec x \cdot y(x))' = \sec x \cdot (2\cos^2 x \sin x - \sec x) \\ = 2\cos x \sin x - \sec^2 x$$

$$\Rightarrow v(x) y(x) = \int 2\cos x \sin x - \sec^2 x \, dx$$

$$\Rightarrow y(x) = \frac{1}{v(x)} \left[\int 2\cos x \sin x - \sec^2 x \, dx \right].$$

Step 1: Standard form!

$$y' = -(\tan x) y + \frac{2\cos^3 x \sin x - 1}{\cos x}.$$

Step 2: Integrating factor! Exponential!

$$\text{Let } v(x) = \exp \left\{ \int (\tan x) \, dx \right\} = \exp \left\{ \ln |\sec x| \right\}$$

$$= |\sec x| \quad (\because x \in [0, \frac{\pi}{2}])$$

$$= \sec x.$$

$$\sec x \cdot y' + \sec x \tan x y$$

$$\therefore v(x) (y' + (\tan x) y) = v(x) \frac{2\cos^3 x \sin x - 1}{\cos x}.$$

$$\therefore [v(x) y]' = \cancel{\sec x} \cdot \frac{2\cos^3 x \sin x - 1}{\cos^2 x} = 2 \cos x \sin x - \sec^2 x.$$

$$\begin{aligned}\therefore v(x) \cdot y &= \int (2 \cos x \sin x - \sec^2 x) dx \\ &= \int \sin 2x dx - \int \sec^2 x dx \\ &= -\frac{1}{2} \cos 2x - \tan x + C.\end{aligned}$$

Step 3: solve $y(x)$!

$$\therefore y(x) = \frac{1}{v(x)} \left[-\frac{1}{2} \cos 2x - \tan x + C \right]$$

$$= \cos x \left[-\frac{1}{2} \cos 2x - \tan x + C \right].$$

□

General solution!

Applications

1. Malthusian Growth Model

If $P = P(t)$ is the population at time t , then

$$\frac{dP(t)}{dt} = k P(t) \quad (\text{separable})$$

Solve for P :

$$\frac{dP(t)}{dt} = k \cdot P(t)$$

$$\Rightarrow \frac{1}{P(t)} dP(t) = k dt$$

$$\Rightarrow \int \frac{1}{P(t)} dP(t) = \int k dt$$

$$\Rightarrow \ln |P(t)| = kt + C_1$$

$$\Rightarrow |P(t)| = \exp \{kt + C_1\}$$

$$\Rightarrow |P(t)| = \exp \{kt\} \cdot C_2 \quad (\text{where } C = e^{C_1} > 0)$$

$$\Rightarrow P(t) = C \cdot \exp \{kt\} \quad (\text{where } C \geq 0)$$

[General solution]

$$P' = kP + 0$$

$$\left| \begin{array}{l} \text{Let } v(x) = \exp \int k dt \\ = e^{-kt} \end{array} \right|$$

$$| e^{-kt} P' - e^{-kt} kP = 0$$

$$\left| [e^{-kt} P(t)]' = 0 \right|$$

$$\left| e^{-kt} P(t) = C \right|$$

$$\left| P(t) = C \cdot e^{kt} \right|$$

If $P(0) = P_0$, then $P_0 = C \cdot e^{k \cdot 0}$. Hence, a particular solution is

$$P(t) = P_0 \cdot e^{kt}$$

where P_0 is the initial population.

2. Logistic Growth Model

In this model, $\frac{dP}{dt} = k \cdot P \cdot \left(1 - \frac{P}{m}\right)$, ($k > 0$)

where $m > 0$ is the carrying capacity constant. Let us assume $0 < P < m$.

Solve for P : $P(t) \equiv 0$ ✓ (assume $0 < P < m$)

$$\frac{1}{P(1 - \frac{P}{m})} dP = k dt$$

$$\frac{1}{P} + \frac{1}{m-P} = \frac{m}{P(m-P)}$$

$$\Rightarrow \int \frac{1}{P(1 - \frac{P}{m})} dP = \int k dt$$

$$\Rightarrow m \int \frac{1}{P(m-P)} dP = kt + C$$

$$\Rightarrow \int \frac{1}{P} + \frac{1}{m-P} dP = kt + C$$

$$\Rightarrow \int \frac{1}{P} dP - \int \frac{1}{P-m} dP = kt + C$$

$$\Rightarrow \ln |P| - \ln |m-P| = kt + C$$

$$\Rightarrow \left| \frac{P}{m-P} \right| = e^{kt+C}$$

$$0 < P < m \Rightarrow \frac{P}{m-P} = e^{kt} \cdot e^{C_1}$$

$$\Rightarrow \frac{m-P}{P} = e^{-kt-C_1} \Rightarrow \frac{m}{P} - 1 = e^{-kt-C_1} \Rightarrow P = \frac{m}{1 + e^{-kt-C_1}} \Rightarrow P = \frac{m}{1 + e^{-kt} \cdot C}$$

The solution $P = \frac{m}{1 + C \cdot e^{-kt}}$ is called the logistic function. If the initial condition is $P(0) = P_0$, then $P_0 = \frac{m}{1+C}$, which implies

$$C = \frac{m}{P_0} - 1$$

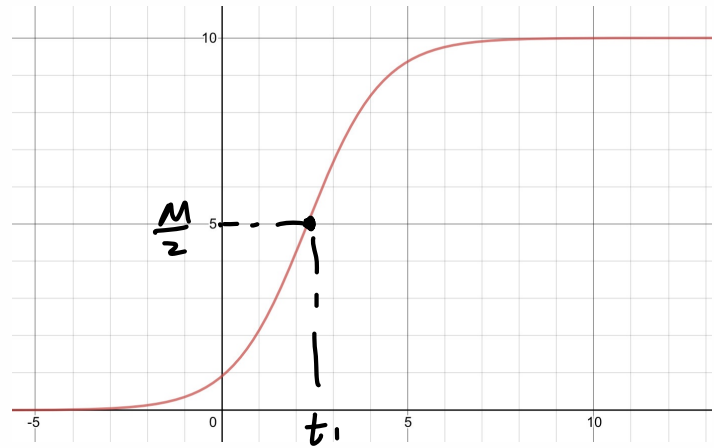
$$\text{and } P(t) = \frac{m}{1 + e^{-kt} \left(\frac{m}{P_0} - 1\right)}$$

Some properties of the logistic function $P = \frac{M}{1 + ce^{-kt}}$.

① P is increasing of t .

② $\lim_{t \rightarrow \infty} P(t) = M$.

③ P has a point of inflection
when its value is $\frac{M}{2}$.



If this point is given by t_1 ,

then P is concave up on $(-\infty, t_1)$
and concave down on (t_1, ∞) .

Applications

Newton's Law of Cooling (or Heating)

Newton's law of cooling (or heating) states that the rate of change of the temperature of an object is proportional to the difference of temperatures between the object and its surroundings. In other words, if $H(t)$ is the temperature of an object at time t , then H satisfies the differential equation

$$\frac{dH}{dt} = k(H - R), \quad k < 0$$

where k is some negative constant (why?) and R is the surrounding temperature, which is a constant.

e.g. The body of a murder victim is found at noon in a room with a constant temperature of 20°C . At noon the temperature of the body is 35°C ; two hours later the temperature of the body is 33°C .

- (a) Find the temperature, H , of the body as a function of t , the time in hours since it was found.
- (b) Assuming that the body had the normal temperature 37°C at the time of murder, estimate the time of the murder.

Sol: (a) Solve an initial value problem:
$$\begin{cases} \frac{dH}{dt} = k(H - 20) \\ H(0) = 35 \end{cases}$$

$$\Rightarrow H' = kH - 20k.$$

$$\Rightarrow \text{Let } v(t) = \exp\left\{\int -k dt\right\} = e^{-kt}.$$

$$\Rightarrow e^{-kt} H' = e^{-kt} (kH - 20k)$$

$$\Rightarrow [V(t)H' - KH V(t)] = -20k V(t)$$

$$\Rightarrow [V(t)H(t)]' = [e^{-kt}H(t)]' = -20k V(t)$$

$$\Rightarrow V(t)H(t) = \int -20k V(t) dt = -20k \int e^{-kt} dt = 20e^{-kt} + C$$

$$\Rightarrow H(t) = e^{kt} [20e^{-kt} + C] = 20 + Ce^{kt}$$

$$\because H(0) = 35 \quad \therefore H(0) = 20 + C \cdot e^0 = 35 \Rightarrow C = 15$$

$$\because H(2) = 33 \quad \therefore 15e^{k \cdot 2} + 20 = 33$$

$$\therefore 2 \cdot k = \ln \frac{13}{15} \quad \Rightarrow k = \frac{1}{2} \ln \frac{13}{15}$$

$$\therefore H(t) = 15 \cdot e^{(\frac{1}{2} \ln \frac{13}{15})t} + 20 = 20 + 15 \cdot \left(\frac{13}{15}\right)^{\frac{t}{2}}$$

$$(b) \text{ Let } H(t) = 37$$

$$\text{Then } H(t) = 20 + 15 \cdot \left(\frac{13}{15}\right)^{\frac{t}{2}} = 37$$

$$\Rightarrow \left(\frac{13}{15}\right)^{\frac{t}{2}} = \frac{17}{15}$$

$$\Rightarrow t = 2 \cdot \frac{\ln \left(\frac{17}{15}\right)}{\ln \left(\frac{13}{15}\right)} \approx \underline{\underline{-1.75}}$$

$\therefore$ Murder happened approximately 1.75 hours before noon.

Q